

Statement of Purpose – Amazon ML Summer School

I became interested in machine learning when I tried to answer a question that really mattered to me: why do customers. What can we do about it. I am a third-year Computer Science student at Vellore Institute of Technology and my grade point average is 8.45. Over the two years I have worked on many projects that involved machine learning, including statistical modeling, deep learning, computer vision and artificial intelligence systems.

My first big project was a Customer Intelligence Platform. I used machine learning to understand customer behavior. The platform used RFM segmentation, customer lifetime value forecasting and purchase propensity to help businesses make decisions. I built the platform using FastAPI. Created a dashboard with Streamlit. This project taught me that machine learning is not about complex algorithms but about making it easy for businesses to use.

I am currently interning at Revalsys Technologies, where I am building a chatbot called DocuMind AI. The chatbot can search for documents and answer questions. I am using FastAPI, ChromaDB and large language models like Gemini and Llama. I have learned how quickly the field of machine learning is changing. It is moving from models to systems that can reason and adapt. I want to learn more about the theory behind machine learning so I can keep up with these changes.

I have also worked on projects that involve using machine learning in a way. For example I built a system to detect hate speech and explain how it works. I used computer vision to detect vehicles and analyze traffic congestion. What I have learned from these projects is that I am interested in problems where the output of the model affects people. I think it is important to make sure the models are fair and robust.

In addition to my work I am the Events Head for IEEE-TEMS at my college. I have helped plan and execute technical events, workshops and hackathons. I have also represented my team at our tech fest. This experience has taught me how to work with teams communicate ideas and meet deadlines. I think these skills are essential for working in the machine learning industry.

I am applying to the Amazon ML Summer School because I want to learn from researchers and practitioners who build machine learning systems. I want to learn more about learning, recommendation systems and reinforcement learning. I want to understand how Amazon balances research ideas with the constraints of production. I hope to share my project experience with peers and mentors and learn from them. I want to leave the program with an understanding of machine learning and a clearer sense of the direction I want to pursue.

If I get the chance to learn from Amazons scientists and engineers I am confident that I will gain skills and a better understanding of how to apply machine learning to real-world problems. I think this will help me make an impact, on peoples lives.